a time series with an additive seasonal pattern

$$y = t + s + e$$

- y → ts
- t → trend
- s → seasonality
- e → to capture the noise/residuals from the model

Two patterns to look for:

1. the trend is approximately Linear
2. diff between the peak & trough for a given year is relatively constant over time

multiplicative model

$$y = TSE$$

Two patterns to look for:

1. exponential (growth) trend in the ts
2. diff between the peaks & trough increase as the values of ts increase

hw:

$$sn_t = \gamma[y_t - L_t] + (1-\gamma)sn_{t-m}$$

estimate for seasonal pattern @ time t

weight/ smoothing parameter

error

weight assigned for your last seasonal factor for this "season/quarter"